Akoizo Original Practice - Form 1 - Answer key

Division C | Mission Possible | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) How many points is each completed scorable action worth?

Unit: Where the points are: building a scoring plan.
Worked-case data: A Regional device starts on the first coin, completes 8 scorable actions and the Final Action at 58 s, earns all ASL points, sets up in 20 minutes, has no adjustments, and measures 60 x 60 x 70 cm. Estimate its score.
Separate investigation observations:
Plan A: 12 actions, works in 6 of 10 test runs.
Plan B: 9 actions, works in 10 of 10 test runs.
Failure cost: A failed run usually loses the Final Action, 75 no-adjust bonus, and some time points.

A. 50

B. 25

C. 100

D. 250

Correct: A

2. (2 points) Target Time 90 s. The run takes 100 s. What are the time points?

A. 200

B. 0

C. 160

D. 180

Correct: C

3. (2 points) What must an endothermic action do to score?

Unit: Chemical actions: gas, cold, and fizz.
Worked-case data: You need about 600 mL of CO_2 to inflate a balloon to size. How much baking soda and vinegar should you use?
Separate investigation observations:
Test 1: 1.0 g baking soda + 30 mL vinegar: balloon expanded 14 cm.
Test 2: 2.0 g + 30 mL vinegar: expanded 21 cm.
Test 3: 3.0 g + 30 mL vinegar: expanded 22 cm; undissolved powder left over.

A. Produce gas

B. Change color

C. Initiate the next action because of the temperature drop

D. Feel cold to the touch

Correct: C

4. (2 points) When does timing begin?

Unit: The Start Action: coins, slots, and selective triggers.
Worked-case data: Design a rail gap that rejects nickels and dimes but passes quarters to a trigger, allowing 0.5 mm tolerance each side.
Separate investigation observations:
Gap 21.5 mm: Nickel sometimes stuck on the rail and triggered the start.
Gap 22.8 mm: Dime and nickel fell through 20 of 20; quarter carried 20 of 20.
Gap 24.0 mm: Quarter fell through in 6 of 20 trials.

A. When the team signals ready

B. When the gum drops

C. When the first action starts

D. When the coin is inserted

Correct: C

5. (2 points) A participant pushes the lever that releases the gum. What is the result?

Unit: The Final Action, the ASL, and competition day.
Worked-case data: Estimate the landing speed of a gum pack dropped 55 cm, and suggest how to keep it on the platform.
Separate investigation observations:
Drop tests: Gum fell 52 cm; landed on platform 7 of 10 times, bouncing off 3.
Change: Added a level release shelf; dropped 51 cm.
New tests: Landed and stayed 10 of 10 times.

A. No Final Action points and zero time points

B. A 25-point touch penalty only

C. Full points

D. Tier 3

Correct: A

6. (4 points) Why should electrical actions be placed early in the sequence?

Unit: Electrical, infrared, and magnetic actions.
Worked-case data: A 9 V battery drives a nichrome segment of 6 Ω. What current and power result, and is the plan legal?
Separate investigation observations:
Timeline: Coin start at 0 s; IR action at 8 s; nichrome cut at 27 s.
Slow test day: Device ran 15 percent slower; cut happened at 31 s.
Supervisor note: Electricity used after 30 s.

answer: Electricity is allowed only in the first 30 seconds of operation, so those actions must finish before then. (4 points).

7. (4 points) Why must the incline and pulley masses be removable?

Unit: Simple machines: levers, inclines, pulleys, and screws.
Worked-case data: A screw has a pitch of 1.5 mm. How many turns move an object 2.0 cm, and is that enough rotation for the action?
Separate investigation observations:
Pulley build: Load 420 g lifts 11 cm; effort rope moves 22 cm.
Supervisor check: Counts two supporting segments.
Team claim: Pulley action meets IMA 3.

answer: So the supervisor can weigh them to confirm they are at least 400 g. (4 points).

8. (4 points) List the four ways to earn ASL points.

Unit: Where the points are: building a scoring plan.
Worked-case data: A Regional device starts on the first coin, completes 8 scorable actions and the Final Action at 58 s, earns all ASL points, sets up in 20 minutes, has no adjustments, and measures 60 x 60 x 70 cm. Estimate its score.
Separate investigation observations:
Plan A: 12 actions, works in 6 of 10 test runs.
Plan B: 9 actions, works in 10 of 10 test runs.
Failure cost: A failed run usually loses the Final Action, 75 no-adjust bonus, and some time points.

answer: Submitted on time; proper format; all actions included and accurately described; device labels match the ASL numbers and task numerals. (4 points).

9. (4 points) Why can’t a scorable action also be the water timer?

Unit: Timing: Target Time, the water timer, and pacing.
Worked-case data: Target Time 60 s. The device runs 67 s with a 25-s water timer that ended at 40 s. Calculate time points and timer bonus.
Separate investigation observations:
Timer calibration: 150 mL -> 18 s; 300 mL -> 30 s; 450 mL -> 39 s.
Other actions: Total 46 s without the timer.
Target Time: 80 s announced at State.

answer: The rules require the water timer to be a separate action that is not one of the scorable actions. (4 points).

10. (4 points) What penalty applies if vinegar splashes out of the device?

Unit: Chemical actions: gas, cold, and fizz.
Worked-case data: You need about 600 mL of CO_2 to inflate a balloon to size. How much baking soda and vinegar should you use?
Separate investigation observations:
Test 1: 1.0 g baking soda + 30 mL vinegar: balloon expanded 14 cm.
Test 2: 2.0 g + 30 mL vinegar: expanded 21 cm.
Test 3: 3.0 g + 30 mL vinegar: expanded 22 cm; undissolved powder left over.

answer: A 50-point penalty for the first solid or liquid leaving the device dimensions. (4 points).

11. (6 points) Solve this case and explain your reasoning: Design a rail gap that rejects nickels and dimes but passes quarters to a trigger, allowing 0.5 mm tolerance each side.

reasoning: Largest coin to reject: nickel, 21.2 mm; gap must be wider than 21.2 + 0.5 = 21.7 mm.
Smallest coin to keep: quarter, 24.3 mm; gap must be narrower than 24.3 - 0.5 = 23.8 mm.
Choose a gap near 22.7 mm, centered in the 21.7-23.8 mm window. (3 points).

conclusion: A gap near 22.7 mm drops both small coins and carries the quarter to the trigger. (3 points).

12. (6 points) Solve this case and explain your reasoning: Estimate the landing speed of a gum pack dropped 55 cm, and suggest how to keep it on the platform.

reasoning: v = sqrt (2gh) = sqrt (2 x 9.8 x 0.55) about 3.3 m/s.
At that speed a tilted pack bounces or slides off a 12 cm platform.
Use a level release guide and center the drop; test slide distance on the real platform. (3 points).

conclusion: Drop flat and centered; adjust the release so the pack lands with room to slide. (3 points).

13. (2 points) What change best protects against the penalty?

Supplied synthetic observations:
Timeline: Coin start at 0 s; IR action at 8 s; nichrome cut at 27 s.
Slow test day: Device ran 15 percent slower; cut happened at 31 s.
Supervisor note: Electricity used after 30 s.

A. Move the nichrome cut earlier in the sequence

B. Use a bigger battery

C. Remove the IR action

Correct: A

14. (2 points) Define “Ideal mechanical advantage (IMA)” in the context of Mission Possible.

answer: Frictionless force multiplication from geometry. (2 points).

15. (2 points) Define “Scorable action” in the context of Mission Possible.

answer: One of 12 listed tasks, 50 points each when completed as described. (2 points).

16. (2 points) Define “Water timer” in the context of Mission Possible.

answer: Designated non-scorable action over 10 s earning 1 pt/s before the Target Time. (2 points).

17. (2 points) Define “Nichrome” in the context of Mission Possible.

answer: Resistance wire that heats with current; used to cut line. (2 points).

18. (2 points) Define “Endothermic” in the context of Mission Possible.

answer: Absorbs heat from the surroundings, causing cooling. (2 points).